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Figure 1: MATLAB script and resulting plot for Channel Capacity vs SNR.

Script (untitled9ii.m):

```
1 clc;
2 clear;
3 close all;
4 snr = 0:5:30;
5 snr_linear = 10.^(snr/10);
6 capacity = log2(1 + snr_linear);
7 figure;
8 plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
9 grid on;
10 xlabel('SNR (dB)');
11 ylabel('Channel Capacity (bps/Hz)');
12 title('Channel Capacity vs SNR');
```

Workspace:

Name	Value
capacity	[1,2.05]
snr	[0,5,10]

Plot: Channel Capacity vs SNR. The x-axis is SNR (dB) ranging from 0 to 30. The y-axis is Channel Capacity (bps/Hz) ranging from 1 to 10. The plot shows a linear relationship between SNR and Channel Capacity, with data points marked by blue circles and connected by a blue line. A grid is visible.

sin_linear [1,3,13] Command Window

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co2_at2.m

practical 1.m

practical_2.m

practical1.m

untitled2.m

untitled2exp

untitled3.m

untitled3a5g

Workspace

Name	Value
antennas	[4,8,16]
gain	[6.020]

```
1 clc;
2 clear;
3 close all;
4 antennas = [4 8 16 32 64]; % Number of Antenna Elements
5 gain = 10*log10(antennas); % Antenna Gain (dB)
6 figure
7 bar(antennas, gain)
8 grid on
9 xlabel('Number of Antenna Elements')
10 ylabel('Antenna Gain (dB)')
11 title('Antenna Gain for Different Antenna Array Sizes')
```

Number of Antenna Elements	Antenna Gain (dB)
4	6.02
8	9.03
16	12.04
32	15.05
64	18.06

Command Window

